Many updates to a resource follow the pattern of first reading the current version and then sending an updated version. You can ensure that nobody has changed the resource since you read it using one of two standard HTTP mechanisms:

- The server can return a Last-Modified header when reading a resource that indicates the date and time when it was last modified. The client can then send an If-Unmodified-Since header in its update request with the same timestamp. If the resource has changed in the meantime, then the request will be rejected with a 412 Precondition Failed status.[2] The main downside of Last-Modified headers is that they are limited to the nearest second, so are unable to detect changes occurring more frequently.
- Alternatively, the server can return an ETag (Entity Tag) header that should change whenever the resource changes as shown in figure 13.6. Typically, the ETag is either a version number or a cryptographic hash of the contents of the resource. The client can then send an If-Matches header containing the expected ETag when it performs an update. If the resource has changed in the meantime, then the ETag will be different and the server will respond with a 412 status-code and reject the request.

  WARNING Although a cryptographic hash can be appealing as an ETag, it does mean that if the ETag will revert to a previous value if the content does. This allows an attacker to replay any old requests with a matching ETag. You can prevent this by including a